

Article

Determination of Wheat Growth Stages Using Image Sequences and Deep Learning

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Abstract: The growth stage of wheat is key information for critical decision-making related to cultivar screening of wheat and farming activities. In order to solve the problem that it is difficult to determine the growth stages of a large number of wheat breeding materials grown in an artificial climate room accurately and quickly, the first attempt was made to determine the growth stages of wheat using image sequences of growth and development. A hybrid model (DenseNet–BiLSTM) based on the DenseNet and Bidirectional Long Short-Term Memory was proposed for determining the growth stage of wheat. The spatiotemporal characteristics of wheat growth and development were modeled by DenseNet–BiLSTM synthetically to classify the growth stage of each wheat image in the sequence. The determination accuracy of the growth stages obtained by the proposed DenseNet–BiLSTM model was 98.43%. Of these, the determination precisions of the tillering, re-greening, jointing, booting, and heading period were 100%, 97.80%, 97.80%, 85.71%, and 95.65%, respectively. In addition, the accurate determination of the growth stages and further analysis of its relationship with meteorological conditions will help biologists, geneticists, and breeders to breed, screen, and evaluate wheat varieties with ecological adaptability.

Keywords: wheat; growth stage determination; deep learning; spatiotemporal information; image sequences



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1. Introduction

The growth stage is a key metric for quantifying the dynamic growth development of wheat, playing a crucial role in selecting superior quality and high-yielding varieties, as well as guiding agricultural practices within the framework of precision agriculture [1–3]. There is a great difference in the demand for water, fertilizer, and pesticide in the different growth stages of wheat. Once the nutrients in the soil are not enough to provide the normal growth of wheat, it will affect the growth and yield of wheat. Even if fertilizer is added at a later stage, it is difficult to recover from the damage caused by passing the corresponding growth period. The determination of growth stages is crucial for precision agriculture to provide theoretical and technical support for the precise management of water and fertilizer [4]. The primary growth stages of wheat encompass the seedling, tillering, re-greening, jointing, booting, heading, flowering, grain-filling, and maturity periods, each of which signifies a distinct and continuous process of growth and development throughout the wheat's phenological progression [5]. The accurate determination of wheat growth stages allowed

for the introduction of uniform coding for each growth stage, which has provided the foundation for the automatic monitoring of wheat growth and development and high-throughput phenotyping platforms. Conventionally, the determination of wheat growth stages is conducted manually through visual inspection [6]. Therefore, the automatic and accurate determination of the growth stages is one of the pending issues to be resolved in precision agriculture.

Computer vision has been recognized as a promising method for determining crop growth stages [6–8]. With the advancement of big data technology and high-performance computing [9,10], machine learning models combined with computer vision have been proposed [11–16]. Those models have opened new avenues for plant phenotyping, enabling the successful extraction, classification, and prediction of plant characteristics and behaviors. On the one hand, many studies of growth stage classification have been conducted using spectral data [17–19]. The growth stages of potato crops were also classified utilizing spectroscopy technology, including the tillering stage, tuber formation stage, tuber bulking stage, and tuber maturation stage [20]. This classification model was established using the support vector machine (SVM) algorithm [21] based on spectral bands and the wavelet coefficients obtained from the continuous wavelet transform of reflectance spectra, demonstrating its strong capability in classifying growth stages of potato crops. On the other hand, numerous articles have focused on classification algorithms of growth stages based on RGB color image analysis. A model of a hybrid tree–fuzzy logic approach was proposed to classify lettuce growth stages by using the classification tree to select the most significant features from the texture features extracted from images [22]. A random forest-based classification model was proposed for identifying rice growth stages based on growth-related factors such as height, canopy cover, and accumulative temperature obtained from image and video data [23]. Maize growth stages were classified based on phenotypic traits and UAV Remote Sensing. Using the random forest regression classifier and combining vegetation index and maize phenotypic traits as inputs for classifying maize growth stages, this method achieved the highest classification accuracy (0.951) [24].

In the above-mentioned studies, artificial information obtained from spectral data and images was used for the determination of wheat growth stages. Monitoring the wheat growth process by machine vision with artificial feature extraction suffered from poor objectivity and low efficiency. In fact, deep learning models have been reported in the application of disease recognition [25,26], phenotype measurement, and crop growth stage detection [27]. A wheat growth stage detection model was proposed based on deep reinforcement learning for using on mobile devices [28]. The wheat growth stage estimation method using convolutional neural networks (CNNs) was proposed for the classification of the growth stage of proximal images [1]. Many CNN models have been used for the determination of crop growth stages. For example, a CNN-based recognition method was proposed for monitoring the wheat growth period. The ResNet50, VGG16, and MobileNet models were used in this method, enabling the precise and efficient automatic recognition of wheat growth stages [29]. A ResNet-50-based method was proposed for rice phenological stage recognition of six rice phenological stages, including seedling, tillering, booting jointing, heading flowering, grain filling, and maturity [30]. Furthermore, a method based on a semantic segmentation network (an improved U-net) was proposed for the recognition of maize growth stages in the field [31]. Using images captured in the field, the recognition accuracy of wheat growth stages based on CNN models has reached 95%. There is still room for improvement in recognition accuracy (by extracting spatiotemporal information additionally). A multi-scale convolutional neural network model, termed MultiScalNet-Wheat (MSN-W), was proposed for identification of wheat growth stages. Multi-scale convolution and attention mechanisms were utilized in this model, enhancing

the ability to learn complex features [32]. Another combined model of convolutional algorithmic structure with self-attention, Maize Hybrid Vision Transformer (MaizeHT), was proposed for maize growth stage recognition [33]. ResNet34 and a Transformer were used in MaizeHT for predicting maize growth stage categories.

As in most related work, deep learning has been applied to the determination of wheat growth stages. In contrast with our work, this has been performed for the determination of wheat growth stages grown in the field. The growth stages of a large amount of germplasm resource materials in an artificial climate room were also still determined manually. The determination of the growth stage manually was labor-intensive and subjective, which are urgent problems to be solved in breeding work. Here, a model was proposed for determining the growth stage of wheat grown in a controlled environment. However, using images captured in a controlled environment, the recognition accuracy of wheat growth stages based on CNN models was as low as 82% tested in this study. CNNs cannot exploit the correlations between successive image frames for properly modeling growth and development dynamic characteristics (spatiotemporal information). The dynamic behaviors during wheat growth and development were also useful for the determination of wheat growth stages. RNNs combined with CNNs have been successfully used to analyze continuous spatiotemporal information [34–38]. The LSTM [39] was a variant of RNN, which made up for the lack of RNN by preserving temporal information over a long time period. One study proposed a complete image processing and machine learning pipeline based on the multiclass CNN, CNN–LSTM, and ConvLSTM to classify three stages of plantlet growth plus soil on the different accessions of two species of red clover and alfalfa. The best classification performance on these types of images was found with the proposed CNN–LSTM model, which achieved 90% accuracy of detection [40]. Therefore, as the higher determination accuracy of crop growth stages was achieved by DenseNet [32], a hybrid model based on the DenseNet and bidirectional LSTM was proposed for determining the growth stage of wheat using image sequences.

This paper addresses the specific challenge of automatically identifying wheat growth stages grown in the controlled environments. To determine the growth stages of a large amount of wheat breeding materials accurately and quickly, the first attempt was made in this study to determine the growth stage using image sequences of wheat growth and development. A hybrid model based on the DenseNet and bidirectional LSTM (DenseNet–BiLSTM) was proposed for determining the growth stage of wheat. The DenseNet and bidirectional LSTM (BiLSTM) in the model were combined to synthetically model the spatiotemporal characteristics of wheat growth and development and classify the growth stage of each wheat image in the sequence. It will also be able to be applied to more crops (such as peanuts, soybeans, etc.) for helping plant biologists, geneticists, and breeders to determine the growth stages accurately. This study's primary contributions are outlined as follows:

- (1) To address the specific issue of automatically recognizing the growth stages of winter wheat grown in controlled environments, a novel approach (DenseNet–BiLSTM) using image sequences of wheat growth and development was proposed.
- (2) The combination of DenseNet and BiLSTM improved the network's capability to learn complex features and improved the extraction of spatiotemporal information from image sequences, enhancing the accuracy of recognizing the growth stages of wheat.

2. Materials and Methods

2.1. Dataset

The determination model (DenseNet–BiLSTM) of wheat growth stages requires a dataset of image sequences and associated ground truth growth stage labels for the purpose of training. In addition, the dataset must cover a large variety of growth stages.

The multi-view dataset of wheat growth and development [41] given by our previous study was used for training and testing the proposed model. The multi-view wheat dataset was comprised of successive images of four different varieties of wheat, which were Fielder, Shannong28 (SN28), Jimai22 (JM22), and Kenong199 (KN199). All wheat samples were cultivated in a growth chamber under controlled conditions (23 °C, 16/8 light/dark, 30 Klux, and 90% humidity). Every pot of wheat was constantly monitored via an image acquisition platform installed at the growth chamber, which contained a camera and rotary table, as shown in Figure 1. The image sequences of wheat growth and development in the multi-view wheat dataset were imaged weekly from eight fixed-side views, as shown in Figure 1. The obtained sequence for each plant from one view involved 6–8 successive images. The number of sequences in the dataset was 592, with a total of 3483 wheat images, including 672 images at tillering period, 1408 images at the re-greening period, 1078 images at the jointing period, 72 images at the booting period, and 253 images at the heading period.

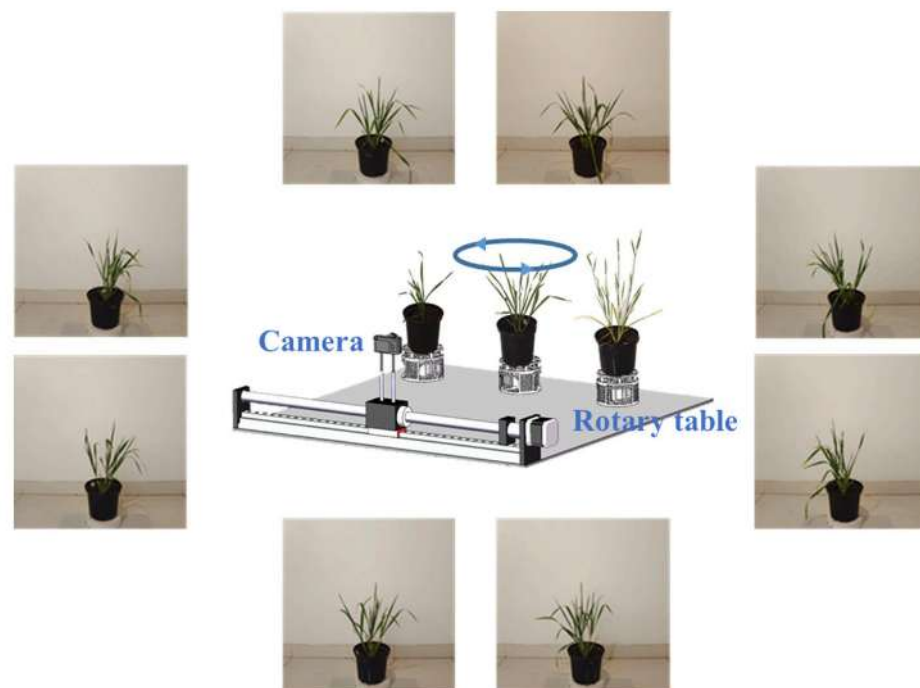


Figure 1. Image acquisition platform and the images obtained weekly from eight fixed-side views.

After collecting successive images of wheat growth and development, images were pre-processed to remove the background. This allowed the determination model DenseNet–BiLSTM to focus on the learning of objective features and avoid the influence of the background on the prediction of growth and development. The wheat image preprocessing was realized using K-means clustering and a color filter, which avoided the large amount of manual annotation time required for deep learning training. Firstly, the wheat image in the RGB color space was converted to the HSV color space. Then, K-means clustering was used to segment the HSV color channel according to the two categories of background and potted wheat. Furthermore, the color threshold segmentation and image cropping of the wheat images in the HSV color space after clustering were carried out to remove the flower

pots and other unremoved backgrounds. Finally, image categories in all wheat growth and development image sequences were manually annotated, including the tillering, re-greening, jointing, booting, and heading period, as shown in Figure 2. The growth stage of wheat images was determined and labeled by an agricultural scientist based on the metrics of the growth stages.

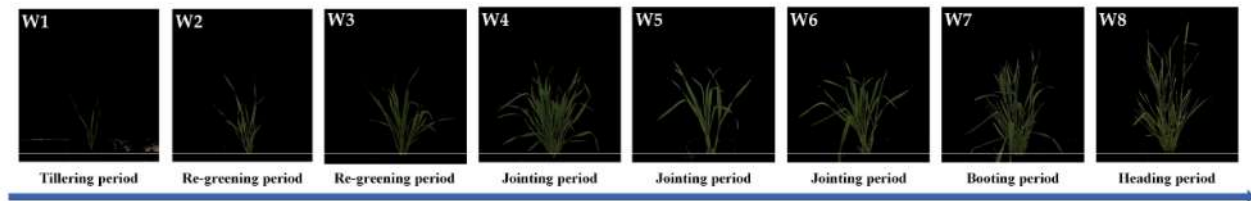


Figure 2. Image sequence of wheat growth and development captured from a consistent perspective.

To avoid cross-sampling, 40 sequences were randomly selected as the test dataset, with a total of 255 wheat images, including 45 images at tillering period, 89 images at the re-greening period, 91 images at the jointing period, 7 images at the booting period, and 23 images at heading period. The other 552 sequences in the dataset were used as the training dataset.

2.2. The Growth Stage Determination Method (DenseNet–BiLSTM) of Wheat Using Image Sequences

The growth stages refer to the continuous growth and development stage of wheat between two adjacent phenological periods. The determination of growth stages was achieved by classifying the growth stages of the wheat images, including the tillering, re-greening, jointing, booting, and heading period. Based on DenseNet and bidirectional LSTMs (BiLSTM), a growth stage determination method (DenseNet–BiLSTM) was proposed for determining wheat growth stages by using image sequences, as shown in Figure 3. The proposed DenseNet–BiLSTM model classified the growth stage of each wheat image in the sequence by synthetically modeling the spatiotemporal characteristics of wheat growth and development (i.e., plant depth visual features and growth dynamic features). The spatiotemporal characteristics of wheat growth and development were modeled by extracting deep visual features from wheat images and temporal dynamics from each direction of the image sequences. The process of the proposed determination method is as shown in Figure 3.

Firstly, the spatial feature vector of each image in the wheat growth and development sequence was sequentially extracted by DenseNet201 and combined into a spatial feature sequence. Secondly, BiLSTM was used to extract the temporal features of spatial feature sequences to model the dynamic behavior of wheat growth and development. The spatiotemporal features of wheat growth and development were modeled by BiLSTM combined with DenseNet. The detailed process of extracting the spatiotemporal characteristics of wheat growth and development is shown as Figure 3. Finally, the temporal feature vectors were passed to a fully connected layer with a Softmax function to determine the growth stage (tillering, re-greening, jointing, booting, and heading period) in each image.

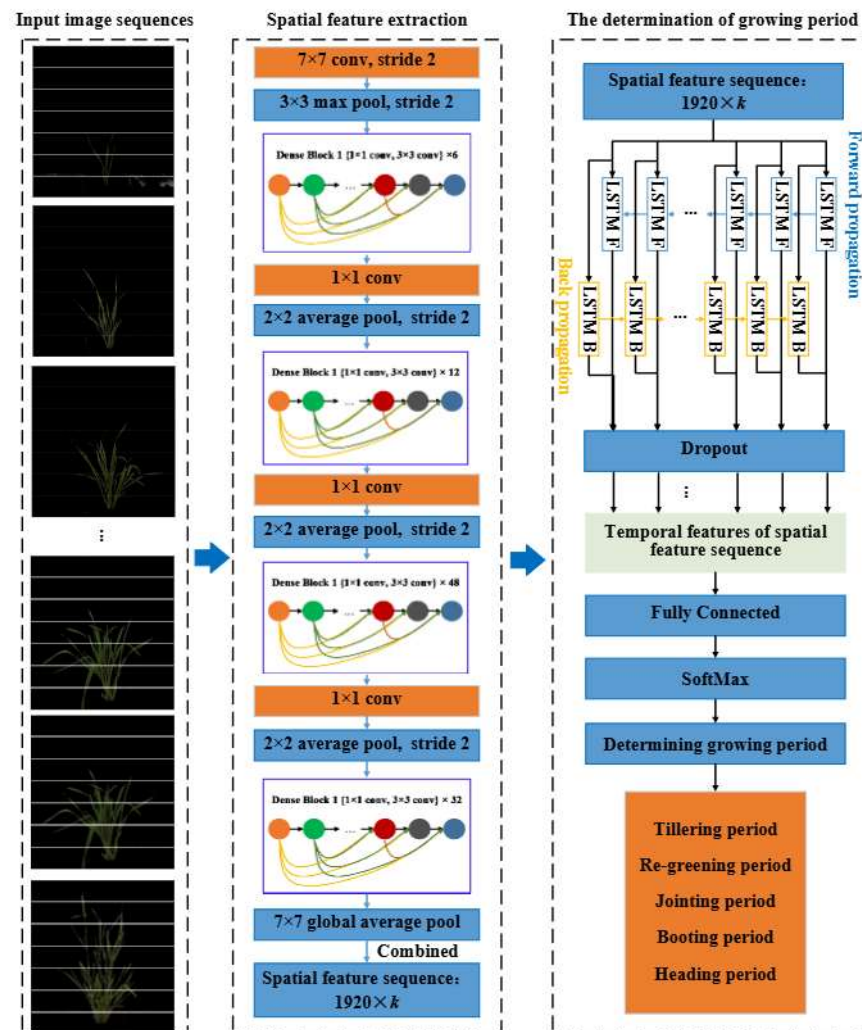


Figure 3. The DenseNet-BiLSTM model for the determination of wheat growth stages. Deep visual features from each individual frame of the wheat image group were extracted using DenseNet. The output fixed-length feature vectors from the images of same sequence were collected as tables of feature vector sequences. BiLSTM was used to extract the temporal features of spatial feature sequences to model the dynamic behavior of wheat growth and development. Illustration of one BiLSTM. “LSTM F” and “LSTM B” are LSTM cells in the forward-time and backward-time directions, respectively. The cells in “LSTM F” share the same parameters, and the cells of “LSTM B” share the same parameters.

2.2.1. Extracting the Spatiotemporal Characteristics of Wheat Growth and Development

In the proposed DenseNet-BiLSTM model, DenseNet with 201 layers (DenseNet201) was used to extract deep visual features of wheat images for modeling the spatial characteristics of wheat growth and development. DenseNet is a CNN model that alleviates the vanishing-gradient problem, strengthens feature propagation, and encourages feature reuse with fewer parameters and less computation [42]. The network structure of DenseNet201 is shown in Figure 3, consisting of convolutional layers, pooling layers, dense block layers, and the fully connected layer. Densely connected blocks were the key to extracting the spatial features of wheat images. The features output from all previous layers were combined and convoluted by the layer of the densely connected block and then the feature output of the current layer was also transmitted to all subsequent layers to promote feature reuse and propagation. A 1×1 convolutional layer was added in front of the convolutional layers of 3×3 to control the number of channels of the feature map and greatly reduce the calculation amount of the model. The output dimension of each densely connected

block was large. As the number of convolutional layer increases, the dimension of the final output feature will be very large. To solve this problem, the downsampling was required between two densely connected blocks. Downsampling usually uses 2×2 pooling layers, but there is also the problem of too many feature map channels resulting in too much computation. As such, the number of channels was reduced to half of the original before downsampling by using a convolutional layer of 1×1 , in which the convolutional layer and the downsampling layer of 1×1 were collectively called transition blocks.

Firstly, 25% of images were randomly selected from the training dataset. The DenseNet201 was pre-trained on those images selected from the training dataset randomly to classify the growth stage of the wheat images. Then, the pre-trained DenseNet201 was used to extract deep visual features from the wheat images of the training dataset. The output features of the same growth sequence were later combined and fed into the BiLSTM layers. BiLSTM was used to extract the temporal features of spatial feature sequences to model the dynamic behavior of wheat growth and development.

The BiLSTM model was composed of multi-bi-directional LSTM layers, including LSTM F cell and LSTM B cell, as shown in Figure 3. In the LSTM cell, an input gate, a forget gate, and an output gate were trainable and are depicted in Figure 4. The writing, reading, and erasing of the information flow in the memory cell were controlled by these gates. The state of the current cell at each time point was computed by analyzing the current input x_t , along with the previous cell output state a_{t-1} and the content of the LSTM memory c_{t-1} , as calculated in Equations (1)–(6). In these Equations (1)–(6), i_t , f_t , and o_t denoted the input gate, forget gate, and output gate, respectively. x was the input of the bi-directional LSTM and y was the output of the bi-directional LSTM (a sequence of hidden vectors). The BiLSTM network was composed of one bi-directional LSTM layer with 2000 hidden units followed by the dropout layer with a probability rate of 0.5 to avoid overfitting.

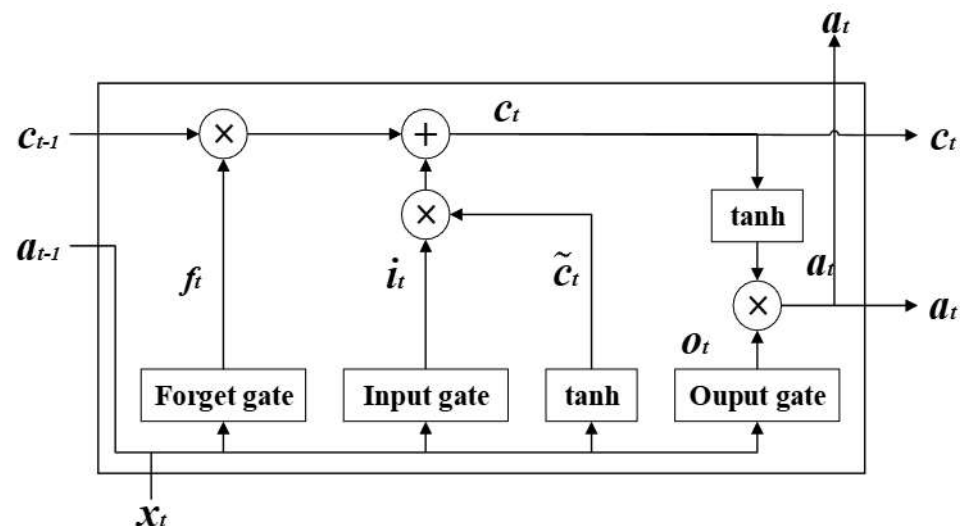


Figure 4. Construction of an LSTM cell. i_t , f_t , and o_t denote the output of input gate, forget gate, and output gate, respectively.

BiLSTM was employed in this proposed DenseNet–BiLSTM model to learn information in both the forward-time and backward-time directions to improve the accuracy of wheat growth stage determination, as shown in Figure 3. The input of the BiLSTM network was feature sequences of plants produced by the DenseNet201 network. BiLSTM was used to extract the temporal features of spatial feature sequences. Then, the spatio-temporal feature modeling of wheat growth and development was modeled by combining DenseNet201 with BiLSTM.

$$\tilde{c}_t = \tanh(W_c[a_{t-1}, x_t] + b_c) \quad (1)$$

$$i_t = \sigma(W_i[a_{t-1}, x_t] + b_i) \quad (2)$$

$$f_t = \sigma(W_f[a_{t-1}, x_t] + b_f) \quad (3)$$

$$o_t = \sigma(W_o[a_{t-1}, x_t] + b_o) \quad (4)$$

$$c_t = i_t \times \tilde{c}_t + f_t \times c_{t-1} \quad (5)$$

$$a_t = o_t \times \tanh(c_t) \quad (6)$$

2.2.2. Experimental Design and Evaluation

To verify the effectiveness of the proposed DenseNet-BiLSTM model for the determination of growth stages, the evaluation precision, recall, and accuracy (Equations (7)–(9)) were used to compare the performance with CNN models (DenseNet201 and AlexNet, VGG, and GoogleNet) and other hybrid models (CNN-LSTM). The growth stages of wheat were classified to verify the effectiveness of the proposed model DenseNet-BiLSTM on the test dataset.

$$\text{sensitivity} = \text{recall} = \frac{TP}{TP + FN} \quad (7)$$

$$\text{precision} = \frac{TP}{TP + TN} \quad (8)$$

$$\text{accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (9)$$

All simulations were performed in MATLAB 2020b. In order to select a suitable network model, the optimal hyperparameters of the proposed DenseNet-BiLSTM model were determined by analyzing the sensitivity of the proposed model to the choice of hyperparameters. Hyperparameters such as the mini-batch size, optimizers, learning rate, and dropout were considered to improve the model performance, as shown in Table 1 and Figure 5.

Table 1. The hyperparameters chosen for the proposed DenseNet-BiLSTM model.

Hyperparameters	Range	Optimal
Learning rate	1×10^{-1} , 1×10^{-2} , 1×10^{-3} , 1×10^{-4} , and 1×10^{-5}	1×10^{-4}
Dropout	0.1, 0.3, 0.5, 0.7, and 0.9	0.5
MiniBatchSize	2, 4, 8, 16, 32, and 64	4
Optimizer	Adam, Sgdm, and RMSProp	Adam

One of the most vital hyperparameters to regulate the deep learning architecture was the optimizer. An optimizer manipulated the value of the weights that minimize the classification error. Three optimizers, Adam, Sgdm, and RMSProp, were considered. As it can be seen from Figure 5, the proposed DenseNet-BiLSTM model with an Adam optimizer performed better than other optimizers. A larger learning rate enhanced the learning process, leading to an increase in the loss function, whereas a small learning rate encouraged the loss function to significantly decrease but increased

the training time. Hence, it was necessary to select the optimal value of the learning rate to minimize the training error, thereby increasing the performance of the model. The proposed DenseNet–BiLSTM model was evaluated with learning rates such as 1×10^{-1} , 1×10^{-2} , 1×10^{-3} , 1×10^{-4} , and 1×10^{-5} . As it can be seen from Figure 5, the proposed DenseNet–BiLSTM model with the learning rate of 1×10^{-4} performed best. Furthermore, the proposed DenseNet–BiLSTM model was evaluated with dropout rates such as 0.1, 0.3, 0.5, 0.7, and 0.9. It was noticed that the proposed model was a little sensitive to the dropout. A small dropout rate caused overfitting and a large dropout rate dramatically hurt performance. The proposed DenseNet–BiLSTM model was also evaluated with MiniBatchSizes such as 4, 8, 16, 32, and 64. A learning rate of 0.0001 was used to train the model for 100 epochs. From Figure 5, it is evident that the accuracy decreased as batch size increased. The classification accuracies of the growth stages, shown in Figure 5, confirmed that using a small MiniBatchSize of 4 achieved optimal results and the MiniBatchSize had little effect on model performance.

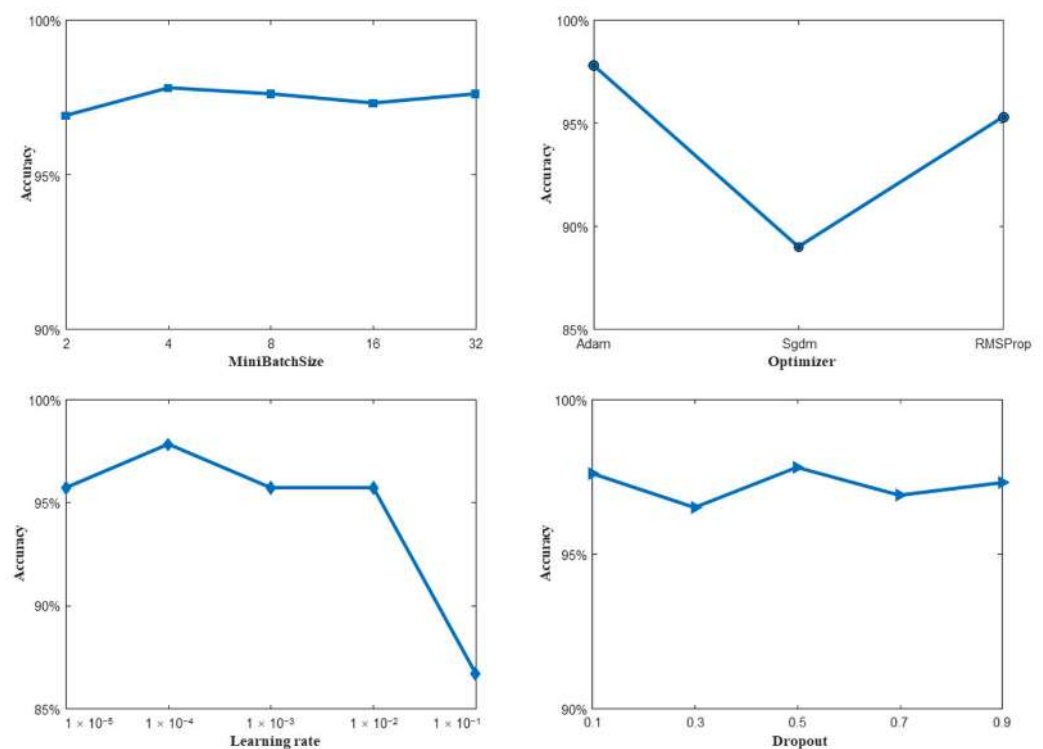


Figure 5. Accuracy comparison between various hyperparameters used in the study.

The proposed DenseNet–BiLSTM model was more sensitive to the learning rate and optimizer. The learning rate and optimizer significantly affected the determination accuracy of the proposed DenseNet–BiLSTM model. MiniBatchSize and dropout had little effect on the determination accuracy of the proposed DenseNet–BiLSTM model, as shown in Figure 5. The best hyperparameters of DenseNet–BiLSTM were identified as follows: a maxepoch of 20, learning rate of 0.0001, dropout of 0.5, and Adam optimizer. The MiniBatchSize was the default value of 4. The best parameters of the proposed model were used in the next experiments and comparisons. The accuracy of the 5-fold cross-validated classification showed that 97.96% of the growth stages were correctly classified by the proposed DenseNet–BiLSTM model.

3. Results

3.1. Experimental Results

When the DenseNet–BiLSTM model was trained, the training and validation data in each round of training were scrambled. The model was validated after each round of training. On the training validation dataset, the loss value curve of the proposed wheat growth stage determination model (DenseNet–BiLSTM) in the training process is shown in Figure 6. The loss decreased to 0 with the increase in the number of iterations and remained stable after 80 iterations. This showed that DenseNet–BiLSTM can effectively learn the spatiotemporal characteristics of wheat growth and development.

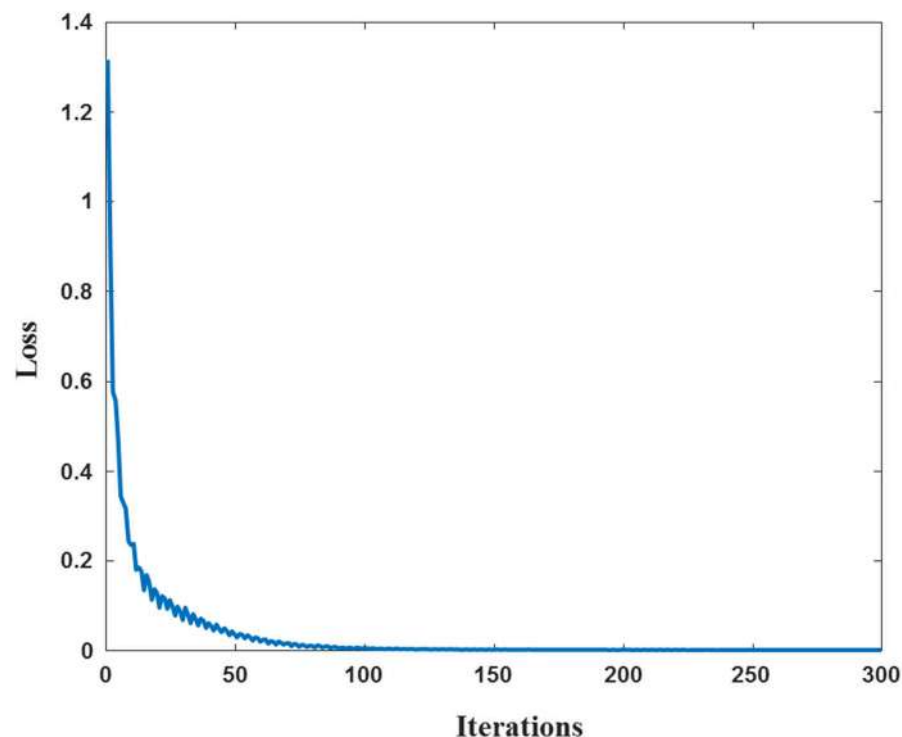


Figure 6. Loss curves of the proposed DenseNet–BiLSTM model in the training process.

The DenseNet–BiLSTM model was tested on the test dataset containing 40 sequences, while maintaining the same training parameters. The determination confusion matrix of growth stages of wheat images in the growth and development image sequences using DenseNet–BiLSTM is shown in Figure 7. The DenseNet–BiLSTM model obtained the determination accuracy of growth stages of 98.43% on the test dataset. The Output Class was the prediction category. The Target Class was the actual category. It can be seen that except for the misjudgment of a few samples, DenseNet–BiLSTM could mostly achieve an accurate determination of the growth stages. The classification recall of the tillering period was 97.78%. The classification recall of the re-greening period was 100%. The classification recall of the jointing period was 98.90%. The classification recall of the booting period was 85.71%. The classification recall of the heading period was 95.65%. The precisions of the determination of tillering and jointing periods were 100%. The determination precision of the re-greening period was 97.80%, the determination precision of the booting period was 85.71%, and the determination precision of the heading period was 95.65%. The precision of wheat in the booting stage was significantly lower than that of other growth stages.

DenseNet-BLSTM Confusion Matrix							
Output Class	Jointing	90 35.3%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	Heading	0 0.0%	22 8.6%	0 0.0%	0 0.0%	1 0.4%	95.7% 4.3%
	Tillering	0 0.0%	0 0.0%	44 17.3%	0 0.0%	0 0.0%	100% 0.0%
	Re-greening	1 0.4%	0 0.0%	1 0.4%	89 34.9%	0 0.0%	97.8% 2.2%
	Booting	0 0.0%	1 0.4%	0 0.0%	0 0.0%	6 2.4%	85.7% 14.3%
		98.9% 1.1%	95.7% 4.3%	97.8% 2.2%	100% 0.0%	85.7% 14.3%	98.4% 1.6%
		Jointing	Heading	Tillering	Re-greening	Booting	
		Target Class					

Figure 7. Confusion matrix of DenseNet-BiLSTM model for determining growth period tested on the test dataset.

3.2. Comparison with Other Models Used in the Existing Classification Methods of Growth Stages

3.2.1. Comparison with CNN Models Used in the Existing Classification Methods

A comparison of the test results of the DenseNet-BiLSTM model and the CNN models used in the existing classification methods of crop growth stages [1] was carried out. The DenseNet201 model and other CNN models (AlexNet, VGG, and GoogleNet) were used to determine the growth stages. Table 2 shows the parameters of the CNN models and their determination accuracy of the growth stages of images. The DenseNet201 used in the proposed model achieved the best growth stage determination accuracy (96.90%) in lower parameters and deeper layers, and the corresponding confusion matrix is shown in Figure 8. The classification recalls of the jointing, heading, tillering, re-greening, and booting periods obtained by the DenseNet201 model were 97.78%, 96.62%, 96.70%, 85.71%, and 100%. This indicated that DenseNet201 has a better ability of feature extraction.

Table 2. The parameters of CNNs and the determination accuracies of growth stages for individual images from the test dataset.

CNNs	Layers	Parameters	Input Size	Feature Vectors Extracted Layer	Output Size	Determination Accuracy of Growth Stages
AlexNet	8	60M	$227 \times 227 \times 3$	Fc7	$1 \times 1 \times 4096$	86.3%
VGG	16	138M	$224 \times 224 \times 3$	Fc7	$1 \times 1 \times 4096$	82.7%
GoogleNet	22	5M	$224 \times 224 \times 3$	Pool5-7 $\times$ 7_s1	$1 \times 1 \times 1024$	85.1%
DenseNet	201	20M	$224 \times 224 \times 3$	Avg_pool	$1 \times 1 \times 1920$	96.86%

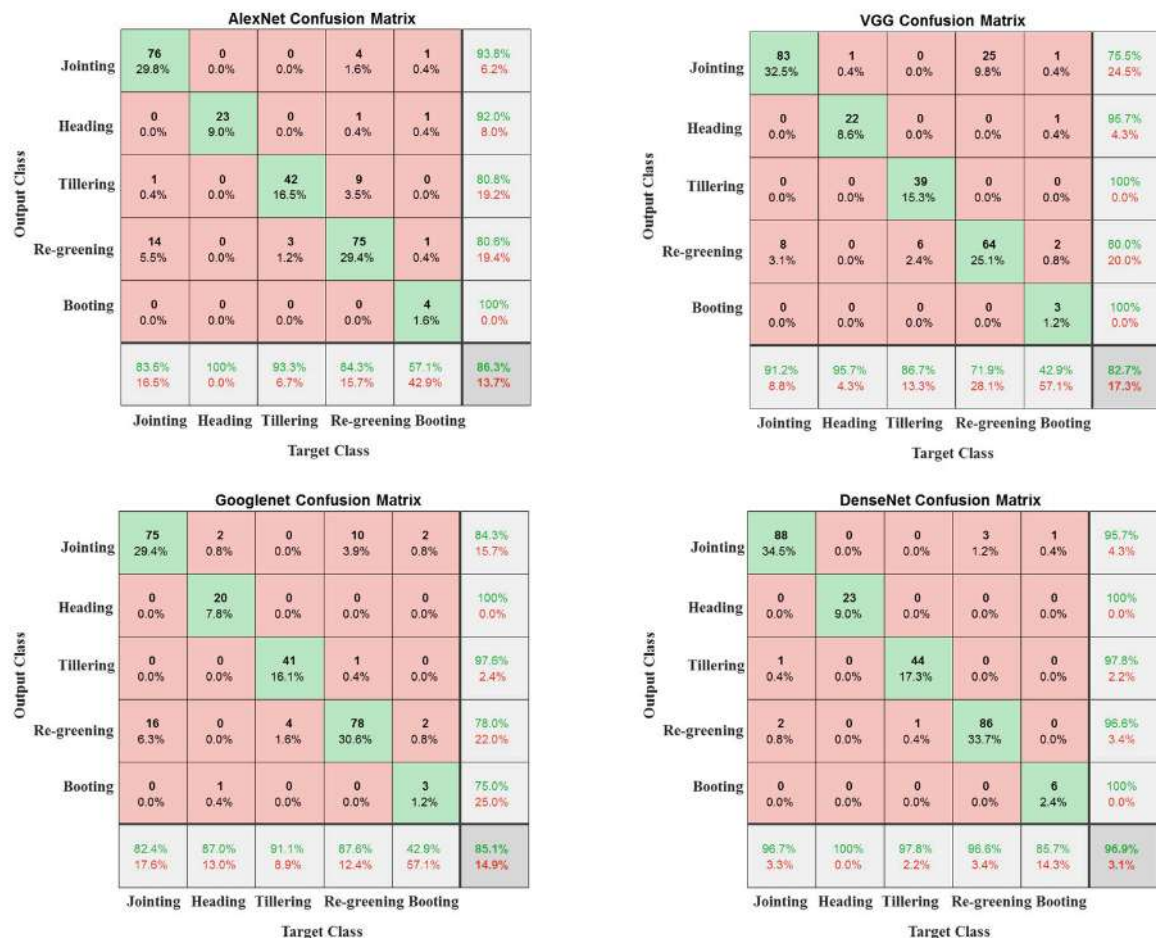


Figure 8. Confusion matrix of different CNN models for determining growth stages tested on the test dataset.

Compared with the DenseNet201 model, the determination accuracy of growth stages obtained by the DenseNet–BiLSTM model was increased by 1.53% and the re-greening period determination recall increased by 3.48%, but the recall rate of the heading period was reduced. The comparison of the test results of the DenseNet–BiLSTM model and the DenseNet201 model verifies that the proposed DenseNet–BiLSTM model has the dual advantages of extracting deep visual features from growth and development sequences and learning temporal features and confirms that considering spatiotemporal features at the same time had great potential and advantages in the determination of growth stages.

Secondly, the DenseNet–BiLSTM, AlexNet–BiLSTM, VGG–BiLSTM, and GoogleNet–BiLSTM models were tested and compared on the test dataset while maintaining the same training parameters. The determination accuracies of growth stages obtained by these models are shown in Table 3. The possibility of the determination of growth stages using image sequences was verified by these results. The determination accuracy of growth stages obtained by the proposed DenseNet–BiLSTM model was significantly higher (7.45%) than that of other models (GoogleNet–BiLSTM and VGG–BiLSTM), as shown in Table 2. Because DenseNet201 mitigates the vanishing gradient problem, strengthens feature propagation, and effectively reuses features, DenseNet201 introduced the direct connections from any layer to all subsequent layers to further increase the diversity of features. As such, richer deep visual features were detected by the Avg_pool layer of DenseNet201 from wheat images, as shown in Table 3.

Table 3. Determination accuracies of growth stages for individual images from the test dataset obtained by different models.

Model	Determination Accuracy of Growth Periods	Determination Precision of Growth Periods	Determination Recall of Growth Periods
AlexNet–BiLSTM	92.94%	89.87%	94.11%
VGG16–BiLSTM	94.12%	93.17%	95.61%
GoogleNet–BiLSTM	90.98%	85.45%	94.15%
DenseNet–BiLSTM	98.43%	95.83%	95.61%

3.2.2. Comparison with Other Hybrid Models Used in the Existing Classification Methods

In addition, under the premise of maintaining the deep visual feature extraction model, the determination accuracies obtained by different CNN–LSTM hybrid models [40] and DenseNet–BiLSTM models on the test set were further compared, as shown in Table 4. The CNN–LSTM hybrid models included DenseNet–LSTM, AlexNet–LSTM, VGG–LSTM, and GoogleNet–LSTM. The DenseNet–BiLSTM model achieved the highest determination accuracy of growth stages compared with different CNN–LSTM hybrid models. Furthermore, the determination accuracy of the growth stages obtained by the network models using BiLSTM to extract temporal features was better than that of the models combined with LSTM. This indicated that DenseNet–BiLSTM could better learn the context information between wheat growth and development image sequences.

Table 4. The determination accuracies of growth stages tested by CNN and LSTM/BiLSTM hybrid models on the test dataset.

Model	Accuracy	Precision	Recall
AlexNet–LSTM(1 Hidden Layer)	88.63%	88.93%	87.04%
AlexNet–LSTM(2 Hidden Layers)	90.20%	85.02%	94.20%
AlexNet–BiLSTM (1 Hidden Layer)	92.94%	89.87%	94.11%
AlexNet–BiLSTM (2 Hidden Layers)	93.33%	94.05%	92.33%
VGG16–LSTM(1 Hidden Layer)	91.76%	89.19%	94.07%
VGG16–LSTM(2 Hidden Layers)	92.16%	92.07%	95.26%
VGG16–BiLSTM (1 Hidden Layer)	94.12%	93.17%	95.61%
VGG16–BiLSTM (2 Hidden Layer)	94.51%	94.03%	96.68%
GoogleNet–LSTM(1 Hidden Layer)	87.45%	85.44%	92.66%
GoogleNet–LSTM(2 Hidden Layers)	88.28%	82.91%	92.64%
GoogleNet–BiLSTM (1 Hidden Layer)	90.98%	85.45%	94.15%
GoogleNet–BiLSTM (2 Hidden Layer)	92.94%	92.48%	96.11%
DenseNet–LSTM (1 Hidden Layer)	94.90%	91.83%	96.12%
DenseNet–LSTM (2 Hidden Layers)	95.69%	94.05%	96.70%
DenseNet–BiLSTM (1 Hidden Layer)	98.43%	95.83%	95.61%
DenseNet–BiLSTM (2 Hidden Layers)	96.86%	92.72%	97.09%

3.3. Comparison with Models with Different Numbers of LSTM/BiLSTM Layers

The determination accuracies obtained by models with different numbers of LSTM/BiLSTM layers were further compared, as shown in Table 4. As shown in Table 4, adding a BiLSTM layer does not necessarily enhance the model's ability to learn more complex sequence features. For the spatial feature sequence extracted by AlexNet, VGG, and GoogleNet, the number of LSTM/BiLSTM layers increased and the accuracy of growth stages increased, as shown in Table 4. For the spatial feature sequence extracted by DenseNet, the accuracy of the determination increased as the number of LSTM layers increased and the determination accuracy decreased slightly as the number of BiLSTM layers increased. These results once again verified that as the depth and width of the model

increased, feature extraction and fusion ability also increased. However, it was not verified that the more complex model provides a better fit.

The determination precisions across each growth stage obtained by models with different numbers of LSTM/BiLSTM layers were compared as shown in Table 5. Misclassifications mostly occurred at the booting stage. This error existed when using the CNN model for the classification of growth and development stages. With the addition of BiLSTM, this problem was solved. The determination precisions of booting stage obtained by VGG16–BiLSTM, GoogleNet–BiLSTM, and DenseNet–BiLSTM were 85.71%. The booting stage was easily misclassified as either the jointing or heading stage. Because of the short booting period and the small number of images in the dataset, those models failed to classify successfully. Compared with the prediction precisions of the re-greening stage using the CNN models, the prediction precisions of the re-greening stage using the hybrid models were significantly improved. This result once again illustrated the effectiveness of considering the spatiotemporal characteristics of wheat growth and development into the classification of wheat growth and development stages.

Table 5. The determination precisions across each growth and development stage tested by CNNs and hybrid models on the test dataset.

Model	Tillering FN	Re-Greening FQ	Jointing BJ	Booting YS	Heading CS
AlexNet–LSTM (1 Hidden Layer)	93.33%	80.90%	93.41%	85.71%	91.30%
AlexNet–LSTM (2 Hidden Layers)	91.11%	94.38%	86.81%	57.14%	90.20%
AlexNet–BiLSTM (1 Hidden Layer)	97.78%	95.51%	89.01%	71.43%	95.65%
AlexNet–BiLSTM (2 Hidden Layers)	91.11%	95.51%	92.31%	100%	91.30%
VGG16–LSTM (1 Hidden Layer)	97.78%	89.89%	91.21%	71.43%	95.65%
VGG16–LSTM (2 Hidden Layers)	97.78%	94.38%	86.81%	85.71%	95.65%
VGG16–BiLSTM (1 Hidden Layer)	97.78%	94.38%	92.31%	85.71%	95.62%
VGG16–BiLSTM (2 Hidden Layer)	97.78%	94.38%	86.81%	85.71%	95.65%
GoogleNet–LSTM (1 Hidden Layer)	91.11%	88.76%	84.62%	71.43%	91.30%
GoogleNet–LSTM (2 Hidden Layers)	93.33%	87.64%	87.91%	50%	95.65%
GoogleNet–BiLSTM (1 Hidden Layer)	91.11%	93.26%	90.11%	57.14%	95.65%
GoogleNet–BiLSTM (2 Hidden Layer)	91.11%	93.26%	92.31%	85.71%	100%
DenseNet–LSTM (1 Hidden Layer)	100%	92.13%	95.6%	71.43%	95.60%
DenseNet–LSTM (2 Hidden Layers)	97.78%	94.38%	96.70%	85.71%	95.65%
DenseNet–BiLSTM (1 Hidden Layer)	100%	97.80%	100%	85.71%	95.65%
DenseNet–BiLSTM (2 Hidden Layers)	97.78%	95.51%	98.90%	71.43%	100%

4. Discussion

4.1. Impact of Different Modules on the DenseNet–BiLSTM Method

In the DenseNet–BiLSTM model, the DenseNet was the heart of the model. As shown in Tables 2 and 3, the DenseNet201 used in the proposed model achieved the best growth stage determination accuracy (96.90%) and the determination accuracy of the proposed model was 98.43%. As shown in Table 4, within a certain number of layers, the better the prediction ability of the CNN models, the better the classification accuracy of models after adding the LSTM/BiLSTM module.

As the depth and width of the model increased, the number of feature extractions and the fusion ability also increased. However, a more complex model was not better, and the model should match the dataset. Overfitting occurred when the model became excessively complex. Conversely, underfitting often occurred when the dataset was more complex and the model was simpler. The layer number of DenseNet was 201, as shown in Table 2. Increasing the number of layers of BiLSTM made the model more complex, but it did not make the prediction accuracy higher. The determination accuracy of the growth stages obtained by the network models using BiLSTM module to extract temporal features was better than that of the models combined with LSTM module. This also showed that the bidirectional mechanism of the BiLSTM module can effectively improve the prediction accuracy of the growth and development stages.

4.2. Highlights of This Study

The comparison of the test results of the DenseNet–BiLSTM model and the DenseNet201 model verified that the proposed DenseNet–BiLSTM model had the dual advantages of extracting deep visual features from growth and development sequences and learning temporal features. This also confirmed that considering spatiotemporal features has great potential and advantages in the determination of growth and development stages. Transformers or attention mechanisms also could be used to capture spatial and temporal variations for the classification of the growth and development stages, such as in references [32,33]. In reference [32], CBAM was added in a feature filter module to enhance the ability to learn complex features for accurately identifying wheat growth stages. The identification of winter wheat growth stages in a complex field environment was 97.6%. ResNet34 and a Transformer were used in reference [33] for predicting maize growth stage categories. However, neither of the above two studies used image sequences. In future work, Transformers or attention mechanisms will be considered for the determination of growth and development stages using image sequences.

This work will lay a technological and theoretical foundation for constructing a wheat growth monitoring system and precise management of wheat [43], such as the precise management of water and fertilizer.

4.3. Limitations and Future Perspectives

In this study, only five growth and development stages of individual wheat plants under controlled conditions were determined. There are still many issues that need to be solved as the model expands beyond the controlled environment, such as light conditions, plant occlusion, and in-field effects. Future work could include stress factors and environmental parameters like soil moisture, temperature, and humidity [44,45] as inputs to make the model adaptable to different climate scenarios. In the future work, stress factors, environmental parameters, and image sequences obtained by using UAVs and the IoT will be used for the determination of the wheat growth and development in complex field environments from 12 stages. Then, this model will be applied for real-time crop monitoring systems in large-scale farming and assessing wheat varieties.

Plant roots are essential for water and nutrient absorption, anchoring, mechanical support, metabolite storage, and interaction with the surrounding soil environment [46,47]. It is necessary to take into account the growth and development of the root system. In the future work, image sequences obtained from wheat root windows will be used for the determination of the growth and development of root systems.

5. Conclusions

The growth stage is a key metric for quantifying the dynamic growth and development of wheat. The growth stage is used as an aid to decision-making for the selection of high-quality and high-yielding wheat varieties and other daily agricultural activities in precision agriculture. In this study, a hybrid DenseNet–BiLSTM model was proposed for determining the growth stage of wheat. DenseNet–BiLSTM modeled the spatiotemporal characteristics of wheat growth and development synthetically and classified the growth stage of each wheat image in the sequence. The determination accuracy of the growth stages was 98.43%. Of these, the determination precisions of the tillering, re-greening, jointing, booting, and heading periods were 100%, 97.80%, 97.80%, 85.71%, and 95.65%, respectively. The test results of the proposed DenseNet–BiLSTM model were also higher compared with the CNN models (AlexNet, VGG, and GoogleNet) and hybrid models (CNN–LSTM) used in the existing classification methods of crop growth stages. This proposed determination method of wheat growth stages verified the possibility of using time series images to determine growth stages. This will also be able to be applied to more crops (such as peanuts, soybeans, etc.) for helping plant biologists, geneticists, and breeders to determine growth stages accurately. In the future, the proposed model will be improved by including meteorological conditions to help breeding experts select, cultivate, screen, and evaluate new wheat varieties with ecological adaptability.

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